

# Chapter Contents

|                                                         |           |
|---------------------------------------------------------|-----------|
| <b>Chapter 15 - Matrices</b>                            | <b>4</b>  |
| <b>Reading Matrix Notation</b>                          | <b>5</b>  |
| The Goal Of This Block                                  | 5         |
| Rows, Columns, And Dimensions                           | 5         |
| Entry Notation                                          | 6         |
| The General Form                                        | 6         |
| When Two Matrices Are Equal                             | 7         |
| Column Vectors And Row Vectors                          | 8         |
| Zero, Diagonal, And Identity Matrices                   | 8         |
| The Transpose                                           | 9         |
| Block 1 Summary                                         | 10        |
| <b>Addition And Scalar Multiplication</b>               | <b>10</b> |
| The Goal Of This Block                                  | 10        |
| The Size Rule For Addition                              | 10        |
| A Complete Addition Example                             | 11        |
| Scalar Multiplication                                   | 11        |
| A Matrix Linear Combination                             | 12        |
| Additive Properties                                     | 13        |
| A Useful Vector Interpretation                          | 14        |
| Block 2 Summary                                         | 15        |
| <b>Matrix Multiplication</b>                            | <b>15</b> |
| The Goal Of This Block                                  | 15        |
| The Inner-Dimension Test                                | 15        |
| Why Rows Meet Columns                                   | 16        |
| A Complete Product Example                              | 17        |
| Changing The Order Changes The Problem                  | 18        |
| Matrix Times Vector                                     | 19        |
| The Column-Combination Interpretation                   | 20        |
| Associativity And Distributivity                        | 21        |
| Block 3 Summary                                         | 22        |
| <b>Identity Matrices, Powers, And Algebraic Caution</b> | <b>23</b> |
| The Goal Of This Block                                  | 23        |
| The Identity Matrix In A Product                        | 23        |
| Powers Of A Square Matrix                               | 24        |

|                                                                 |           |
|-----------------------------------------------------------------|-----------|
| A Complete Power Example . . . . .                              | 24        |
| Matrix Squaring Is Not Entrywise Squaring . . . . .             | 25        |
| Expanding A Matrix Square . . . . .                             | 26        |
| Nonzero Matrices Can Multiply To Zero . . . . .                 | 26        |
| Block 4 Summary . . . . .                                       | 27        |
| <b>Matrices Whose Entries Are Functions . . . . .</b>           | <b>28</b> |
| The Goal Of This Block . . . . .                                | 28        |
| A Matrix-Valued Function . . . . .                              | 28        |
| Differentiate Entry By Entry . . . . .                          | 28        |
| Integrate Entry By Entry . . . . .                              | 29        |
| The Matrix Product Rule . . . . .                               | 30        |
| A Complete Product-Rule Check . . . . .                         | 30        |
| A Matrix Initial-Value Calculation . . . . .                    | 32        |
| Block 5 Summary . . . . .                                       | 33        |
| <b>Determinants, Eigenvalues, And Eigenvectors . . . . .</b>    | <b>34</b> |
| The Goal Of This Block . . . . .                                | 34        |
| The Two-By-Two Determinant . . . . .                            | 34        |
| A Three-By-Three Determinant . . . . .                          | 35        |
| The Characteristic Polynomial . . . . .                         | 36        |
| A Complete Two-By-Two Eigenvalue Example . . . . .              | 37        |
| What An Eigenvector Adds . . . . .                              | 38        |
| Finding The Eigenvectors In The Example . . . . .               | 39        |
| Why Eigenvalues Matter For Differential Equations . . . . .     | 41        |
| Triangular Matrices And Multiplicity . . . . .                  | 42        |
| Block 6 Summary . . . . .                                       | 43        |
| <b>Cayley-Hamilton And A Reliable Matrix Workflow . . . . .</b> | <b>44</b> |
| The Goal Of This Block . . . . .                                | 44        |
| Substituting A Matrix Into A Polynomial . . . . .               | 44        |
| The Cayley-Hamilton Theorem . . . . .                           | 45        |
| A Complete Cayley-Hamilton Verification . . . . .               | 45        |
| Reducing A High Matrix Power . . . . .                          | 47        |
| A Reliable Workflow . . . . .                                   | 49        |
| Common Mistake: Multiplying Corresponding Entries . . . . .     | 50        |
| Common Mistake: Checking The Outer Dimensions . . . . .         | 50        |
| Common Mistake: Reversing A Product . . . . .                   | 51        |
| Common Mistake: Losing Constants In A Matrix Integral . . . . . | 51        |

|                                                                                   |           |
|-----------------------------------------------------------------------------------|-----------|
| Common Mistake: Treating An Eigenvalue As A Vector . . . . .                      | 51        |
| Common Mistake: Reading Multiplicity From A Matrix Entry . . .                    | 52        |
| Common Mistake: Substituting A Scalar Constant Into Cayley-<br>Hamilton . . . . . | 52        |
| Block 7 Summary . . . . .                                                         | 52        |
| <b>Original Practice Set . . . . .</b>                                            | <b>53</b> |
| Practice Problems . . . . .                                                       | 53        |
| Worked Answers: Problems 1 To 4 . . . . .                                         | 57        |
| Worked Answers: Problems 5 To 8 . . . . .                                         | 63        |
| Worked Answers: Problems 9 To 12 . . . . .                                        | 69        |
| Chapter 15 Final Summary . . . . .                                                | 77        |

# Chapter 15 - Matrices

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The preceding chapters studied one unknown function at a time. Later chapters will place several unknown functions into a single vector and write their differential equations as one matrix equation.

A matrix is more than a rectangular storage box. Its dimensions determine which operations are legal, and its multiplication rule describes how several quantities interact.

The main questions are:

- how are rows, columns, entries, and vectors described precisely?
- when can matrices be added or multiplied?
- why is matrix multiplication based on rows and columns?
- why can changing the order of a product change the answer?
- how are powers and the identity matrix defined?
- how are matrices containing functions differentiated and integrated?
- how does a determinant produce a characteristic polynomial?
- what do eigenvalues and eigenvectors say about a matrix?
- how does the Cayley-Hamilton theorem reduce high matrix powers?

## Reading Matrix Notation

### The Goal Of This Block

The goal is to read a matrix unambiguously. Before performing an operation, the reader should be able to state the matrix's size and locate any requested entry.

### Rows, Columns, And Dimensions

A matrix is a rectangular arrangement of entries. For example:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

Matrix  $A$  has:

- two horizontal rows
- three vertical columns

Its dimensions are therefore:

$$\boxed{2 \times 3}.$$

The order matters. A  $2 \times 3$  matrix has two rows and three columns, not three rows and two columns.

The practical meaning:

**dimensions are the first compatibility check for every matrix operation.**

## Entry Notation

For a matrix  $A$ , the symbol  $a_{ij}$  denotes the entry in:

row  $i$ ,      column  $j$ .

Return to:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

The entry in row 1, column 3 is:

$$\boxed{a_{13} = 4}.$$

The entry in row 2, column 1 is:

$$\boxed{a_{21} = 0}.$$

The first index always names the row. The second index always names the column.

## The General Form

An  $m \times n$  matrix has  $m$  rows and  $n$  columns:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}.$$

Here:

- $m$  counts rows
- $n$  counts columns
- $a_{ij}$  is the entry in row  $i$ , column  $j$

If  $m = n$ , the matrix is **square**. A square matrix has the same number of rows and columns.

### When Two Matrices Are Equal

Two matrices are equal only when both conditions hold:

1. they have the same dimensions;
2. every pair of corresponding entries is equal.

For example:

$$\begin{bmatrix} x+1 & 4 \\ -2 & y \end{bmatrix} = \begin{bmatrix} 6 & 4 \\ -2 & 3 \end{bmatrix}.$$

Equality of the top-left entries gives:

$$x + 1 = 6.$$

Subtract 1 from both sides:

$$x = 5.$$

Equality of the bottom-right entries gives:

$$y = 3.$$

Therefore the matrices are equal when:

$$\boxed{x = 5, \quad y = 3}.$$

## Column Vectors And Row Vectors

A matrix with one column is a **column vector**:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}.$$

Its dimensions are  $3 \times 1$ .

A matrix with one row is a **row vector**:

$$\mathbf{r} = \begin{bmatrix} r_1 & r_2 & r_3 \end{bmatrix}.$$

Its dimensions are  $1 \times 3$ .

These contain the same number of entries, but they do not have the same shape:

$$3 \times 1 \neq 1 \times 3.$$

In systems of differential equations, unknowns are usually collected into a column vector such as:

$$\mathbf{x}(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \end{bmatrix}.$$

## Zero, Diagonal, And Identity Matrices

A **zero matrix** has zero in every position. Its dimensions must still be specified. For example:

$$0_{2 \times 3} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$



A **diagonal matrix** is square and has zero away from its main diagonal:

$$D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 7 \end{bmatrix}.$$

The **identity matrix** has 1 on the main diagonal and 0 elsewhere:

$$I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Its role in multiplication is similar to the role of the number 1 in ordinary arithmetic. That property will be verified after matrix multiplication has been defined.

### The Transpose

The transpose of  $A$ , written  $A^T$ , turns rows into columns.

Start with:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}.$$

The first row of  $A$  becomes the first column of  $A^T$ , and the second row becomes the second column:

$$A^T = \begin{bmatrix} 2 & 0 \\ -1 & 3 \\ 4 & 5 \end{bmatrix}.$$

Thus a  $2 \times 3$  matrix becomes a  $3 \times 2$  matrix.

## Block 1 Summary

For an  $m \times n$  matrix:

$$a_{ij} = \text{entry in row } i \text{ and column } j.$$

Dimensions are written as rows by columns. Vectors are one-row or one-column matrices, square matrices have equal row and column counts, and transposition interchanges rows with columns.

## Addition And Scalar Multiplication

### The Goal Of This Block

The goal is to combine matrices entry by entry and to distinguish this process from matrix multiplication.

### The Size Rule For Addition

Matrices can be added only when they have the same dimensions.

If:

$$A = [a_{ij}], \quad B = [b_{ij}]$$

are both  $m \times n$ , then:

$$\boxed{A + B = [a_{ij} + b_{ij}]}$$

Each entry is added only to the entry occupying the same position.

A  $2 \times 3$  matrix cannot be added to a  $3 \times 2$  matrix. Having the same number of entries is not enough; the shapes must match.

### A Complete Addition Example

Let:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 6 & 1 \\ 7 & -2 & 0 \end{bmatrix}.$$

Both matrices are  $2 \times 3$ , so  $A + B$  is defined and will also be  $2 \times 3$ .

Add corresponding entries:

$$\begin{aligned} A + B &= \begin{bmatrix} 2 + (-3) & -1 + 6 & 4 + 1 \\ 0 + 7 & 3 + (-2) & 5 + 0 \end{bmatrix} \\ &= \begin{bmatrix} -1 & 5 & 5 \\ 7 & 1 & 5 \end{bmatrix}. \end{aligned}$$

Therefore:

$$A + B = \begin{bmatrix} -1 & 5 & 5 \\ 7 & 1 & 5 \end{bmatrix}.$$

### Scalar Multiplication

A scalar is a single number or scalar-valued function. Multiplying a matrix by a scalar multiplies every entry.

Using:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix},$$

multiply every entry by 3:

$$\begin{aligned} 3A &= \begin{bmatrix} 3(2) & 3(-1) & 3(4) \\ 3(0) & 3(3) & 3(5) \end{bmatrix} \\ &= \begin{bmatrix} 6 & -3 & 12 \\ 0 & 9 & 15 \end{bmatrix}. \end{aligned}$$

Scalar multiplication does not change the matrix dimensions.

### A Matrix Linear Combination

Continue with:

$$A = \begin{bmatrix} 2 & -1 & 4 \\ 0 & 3 & 5 \end{bmatrix}, \quad B = \begin{bmatrix} -3 & 6 & 1 \\ 7 & -2 & 0 \end{bmatrix}.$$

Find  $3A - 2B$ .

First calculate  $3A$ :

$$3A = \begin{bmatrix} 6 & -3 & 12 \\ 0 & 9 & 15 \end{bmatrix}.$$

Now calculate  $2B$ :

$$2B = \begin{bmatrix} -6 & 12 & 2 \\ 14 & -4 & 0 \end{bmatrix}.$$

Subtract corresponding entries:

$$\begin{aligned} 3A - 2B &= \begin{bmatrix} 6 - (-6) & -3 - 12 & 12 - 2 \\ 0 - 14 & 9 - (-4) & 15 - 0 \end{bmatrix} \\ &= \begin{bmatrix} 12 & -15 & 10 \\ -14 & 13 & 15 \end{bmatrix}. \end{aligned}$$

The completed linear combination is:

$$3A - 2B = \begin{bmatrix} 12 & -15 & 10 \\ -14 & 13 & 15 \end{bmatrix}.$$

### Additive Properties

For matrices of compatible dimensions:

$$A + B = B + A$$

and:

$$(A + B) + C = A + (B + C).$$

The zero matrix is the additive identity:

$$A + 0 = A.$$

The matrix  $-A$  is formed by negating every entry, so:

$$A + (-A) = 0.$$

These properties work because matrix addition is ordinary addition performed independently in each position.

## A Useful Vector Interpretation

Let:

$$\mathbf{u} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 5 \\ -1 \end{bmatrix}.$$

Both are  $3 \times 1$ , so  $2\mathbf{u} - \mathbf{v}$  is defined.

First:

$$2\mathbf{u} = \begin{bmatrix} 2 \\ -4 \\ 8 \end{bmatrix}.$$

Then:

$$\begin{aligned} 2\mathbf{u} - \mathbf{v} &= \begin{bmatrix} 2 - 3 \\ -4 - 5 \\ 8 - (-1) \end{bmatrix} \\ &= \begin{bmatrix} -1 \\ -9 \\ 9 \end{bmatrix}. \end{aligned}$$

What this tells us:

**vector linear combinations use the same entrywise addition and scalar multiplication rules as all other matrices.**

## Block 2 Summary

Addition requires equal dimensions:

$$A + B = [a_{ij} + b_{ij}].$$

Scalar multiplication affects every entry:

$$\alpha A = [\alpha a_{ij}].$$

Both operations preserve the dimensions of the matrices involved.

## Matrix Multiplication

### The Goal Of This Block

The goal is to make row-by-column multiplication predictable rather than mysterious. The dimension test will be applied before any arithmetic begins.

### The Inner-Dimension Test

Suppose:

$$A \text{ is } m \times n$$

and:

$$B \text{ is } n \times p.$$

Then  $AB$  is defined because the two inner dimensions match:

$$(m \times \boxed{n})(\boxed{n} \times p).$$

The product has the two outer dimensions:

$$\boxed{AB \text{ is } m \times p}.$$

If the inner dimensions do not match, the product is not defined.

### Why Rows Meet Columns

Let  $A = [a_{ij}]$  be  $m \times n$  and let  $B = [b_{ij}]$  be  $n \times p$ . The entry in row  $i$ , column  $j$  of  $AB$  is:

$$(AB)_{ij} = \sum_{k=1}^n a_{ik}b_{kj}.$$

This means:

1. take row  $i$  from  $A$ ;
2. take column  $j$  from  $B$ ;
3. multiply corresponding entries;
4. add those products.

The matching inner dimension ensures that the selected row and column contain the same number of entries.



## A Complete Product Example

Let:

$$A = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 3 & 2 \end{bmatrix}.$$

The dimensions are:

$$A : 2 \times 3, \quad B : 3 \times 2.$$

The inner dimensions match, so:

$$AB \text{ is } 2 \times 2.$$

Find the row 1, column 1 entry:

$$\begin{aligned} (AB)_{11} &= \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\ &= 1(2) + 2(-1) + (-1)(3) \\ &= 2 - 2 - 3 \\ &= -3. \end{aligned}$$

Find the row 1, column 2 entry:

$$\begin{aligned} (AB)_{12} &= \begin{bmatrix} 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \\ &= 1(1) + 2(5) + (-1)(2) \\ &= 1 + 10 - 2 \\ &= 9. \end{aligned}$$

Find the row 2, column 1 entry:

$$\begin{aligned}(AB)_{21} &= \begin{bmatrix} 3 & 0 & 4 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\ &= 3(2) + 0(-1) + 4(3) \\ &= 6 + 0 + 12 \\ &= 18.\end{aligned}$$

Find the row 2, column 2 entry:

$$\begin{aligned}(AB)_{22} &= \begin{bmatrix} 3 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 5 \\ 2 \end{bmatrix} \\ &= 3(1) + 0(5) + 4(2) \\ &= 3 + 0 + 8 \\ &= 11.\end{aligned}$$

Place each result in its matching position:

$$AB = \begin{bmatrix} -3 & 9 \\ 18 & 11 \end{bmatrix}.$$

### Changing The Order Changes The Problem

For the same matrices:

$$A : 2 \times 3, \quad B : 3 \times 2,$$

the reverse product has dimensions:

$$(3 \times 2)(2 \times 3).$$

Therefore  $BA$  is defined, but it is  $3 \times 3$ , not  $2 \times 2$ .

Compute it row by row:

$$\begin{aligned} BA &= \begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 3 & 0 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 2(1) + 1(3) & 2(2) + 1(0) & 2(-1) + 1(4) \\ -1(1) + 5(3) & -1(2) + 5(0) & -1(-1) + 5(4) \\ 3(1) + 2(3) & 3(2) + 2(0) & 3(-1) + 2(4) \end{bmatrix} \\ &= \begin{bmatrix} 5 & 4 & 2 \\ 14 & -2 & 21 \\ 9 & 6 & 5 \end{bmatrix}. \end{aligned}$$

Here  $AB$  and  $BA$  do not even have the same dimensions. In general:

$$\boxed{AB \neq BA}.$$

Matrix multiplication is not commutative.

### Matrix Times Vector

Let:

$$M = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 4 \\ -2 \end{bmatrix}.$$

The dimensions are:

$$(2 \times 2)(2 \times 1),$$

so  $M\mathbf{x}$  is a  $2 \times 1$  vector.

Use one row of  $M$  at a time:

$$\begin{aligned} M\mathbf{x} &= \begin{bmatrix} 2(4) + (-1)(-2) \\ 1(4) + 3(-2) \end{bmatrix} \\ &= \begin{bmatrix} 8 + 2 \\ 4 - 6 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ -2 \end{bmatrix}. \end{aligned}$$

Thus:

$$M\mathbf{x} = \begin{bmatrix} 10 \\ -2 \end{bmatrix}.$$

### The Column-Combination Interpretation

The same product can be understood using the columns of  $M$ .

Write:

$$M = \begin{bmatrix} | & | \\ \mathbf{m}_1 & \mathbf{m}_2 \\ | & | \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then:

$$M\mathbf{x} = x_1\mathbf{m}_1 + x_2\mathbf{m}_2.$$

For the previous example:

$$\mathbf{m}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{m}_2 = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \quad x_1 = 4, \quad x_2 = -2.$$

Substitute:

$$\begin{aligned} M\mathbf{x} &= 4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 2 \begin{bmatrix} -1 \\ 3 \end{bmatrix} \\ &= \begin{bmatrix} 8 \\ 4 \end{bmatrix} + \begin{bmatrix} 2 \\ -6 \end{bmatrix} \\ &= \begin{bmatrix} 10 \\ -2 \end{bmatrix}. \end{aligned}$$

Another way to see it:

**multiplying by a matrix forms a linear combination of that matrix's columns, using the input vector entries as coefficients.**

### Associativity And Distributivity

Whenever the dimensions make all products meaningful:

$$A(BC) = (AB)C.$$

Thus matrix multiplication is associative.

It also distributes over addition:

$$A(B + C) = AB + AC$$

and:

$$(A + B)C = AC + BC.$$

The order of factors has not been changed in any of these rules.

### Block 3 Summary

The product:

$$(m \times n)(n \times p)$$

is defined and has dimensions:

$$m \times p.$$

Each product entry is a row-column dot product. Matrix multiplication is associative and distributive, but it is not generally commutative.

## Identity Matrices, Powers, And Algebraic Caution

### The Goal Of This Block

The goal is to define powers of square matrices and identify which familiar algebra rules survive matrix multiplication.

### The Identity Matrix In A Product

For an  $m \times n$  matrix  $A$ :

$$I_m A = A$$

and:

$$A I_n = A.$$

The identity on the left must have  $m$  rows and columns. The identity on the right must have  $n$  rows and columns.

For a square  $2 \times 2$  matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Multiply  $A$  by  $I_2$ :

$$\begin{aligned} A I_2 &= \begin{bmatrix} a(1) + b(0) & a(0) + b(1) \\ c(1) + d(0) & c(0) + d(1) \end{bmatrix} \\ &= \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ &= A. \end{aligned}$$

Multiplication by the identity leaves the matrix unchanged.

## Powers Of A Square Matrix

If  $A$  is square and  $n$  is a positive integer, define:

$$A^n = \underbrace{AA \cdots A}_{n \text{ factors}}.$$

In particular:

$$A^2 = AA, \quad A^3 = AAA.$$

The zeroth power is defined by:

$$A^0 = I.$$

A nonsquare matrix does not have an ordinary square  $A^2$ , because the inner dimensions of  $AA$  would not match.

## A Complete Power Example

Let:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}.$$

First calculate  $A^2 = AA$ :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(0) & 1(2) + 2(3) \\ 0(1) + 3(0) & 0(2) + 3(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}. \end{aligned}$$



Now use  $A^3 = A^2A$ :

$$\begin{aligned} A^3 &= \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 8(0) & 1(2) + 8(3) \\ 0(1) + 9(0) & 0(2) + 9(3) \end{bmatrix} \\ &= \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}. \end{aligned}$$

The resulting third power is:

$$A^3 = \begin{bmatrix} 1 & 26 \\ 0 & 27 \end{bmatrix}.$$

### Matrix Squaring Is Not Entrywise Squaring

For:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix},$$

the actual matrix square is:

$$A^2 = \begin{bmatrix} 1 & 8 \\ 0 & 9 \end{bmatrix}.$$

Squaring each entry would instead give:

$$\begin{bmatrix} 1^2 & 2^2 \\ 0^2 & 3^2 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 9 \end{bmatrix},$$

which is different.

What to remember:

**the exponent in  $A^2$  means the matrix product  $AA$ , not entrywise exponentiation.**

### Expanding A Matrix Square

Start from:

$$(A + B)^2 = (A + B)(A + B).$$

Distribute without changing the order of any factors:

$$\begin{aligned}(A + B)^2 &= A(A + B) + B(A + B) \\ &= A^2 + AB + BA + B^2.\end{aligned}$$

The correct matrix expansion is:

$$(A + B)^2 = A^2 + AB + BA + B^2.$$

For ordinary numbers,  $AB = BA$ , so the middle terms combine to  $2AB$ . For matrices, that combination is valid only when  $A$  and  $B$  commute.

### Nonzero Matrices Can Multiply To Zero

Let:

$$P = \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}, \quad Q = \begin{bmatrix} 2 & 0 \\ 2 & 0 \end{bmatrix}.$$

Neither matrix is the zero matrix. Calculate  $PQ$ :

$$\begin{aligned} PQ &= \begin{bmatrix} 1(2) + (-1)(2) & 1(0) + (-1)(0) \\ 1(2) + (-1)(2) & 1(0) + (-1)(0) \end{bmatrix} \\ &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \end{aligned}$$

This calculation shows that two nonzero matrices can have a zero product:

$$\boxed{P \neq 0, \quad Q \neq 0, \quad \text{but} \quad PQ = 0}.$$

This is one reason scalar cancellation habits cannot be imported blindly into matrix algebra.

### Block 4 Summary

For square matrices:

$$A^0 = I, \quad A^n = \underbrace{AA \cdots A}_{n \text{ factors}}.$$

Matrix powers use matrix multiplication. Factor order matters,  $(A + B)^2$  contains both  $AB$  and  $BA$ , and a product of nonzero matrices can equal the zero matrix.

## Matrices Whose Entries Are Functions

### The Goal Of This Block

The goal is to differentiate and integrate matrices entry by entry while preserving the order of factors in product rules.

### A Matrix-Valued Function

A matrix may depend on the independent variable  $t$ . For example:

$$A(t) = \begin{bmatrix} t^2 & e^t \\ \sin t & 3t \end{bmatrix}.$$

Each position contains a scalar-valued function of  $t$ .

The matrix has fixed dimensions  $2 \times 2$ , even though its entries change with time.

### Differentiate Entry By Entry

For:

$$A(t) = [a_{ij}(t)],$$

define:

$$\boxed{A'(t) = [a'_{ij}(t)]}.$$

Using:

$$A(t) = \begin{bmatrix} t^2 & e^t \\ \sin t & 3t \end{bmatrix},$$

differentiate each entry in its current position:

$$\begin{aligned} A'(t) &= \begin{bmatrix} \frac{d}{dt}(t^2) & \frac{d}{dt}(e^t) \\ \frac{d}{dt}(\sin t) & \frac{d}{dt}(3t) \end{bmatrix} \\ &= \begin{bmatrix} 2t & e^t \\ \cos t & 3 \end{bmatrix}. \end{aligned}$$

Thus:

$$A'(t) = \begin{bmatrix} 2t & e^t \\ \cos t & 3 \end{bmatrix}.$$

### Integrate Entry By Entry

An indefinite matrix integral is also calculated entry by entry:

$$\int A(t) dt = \left[ \int a_{ij}(t) dt \right].$$

For the same  $A(t)$ :

$$\begin{aligned} \int A(t) dt &= \begin{bmatrix} \int t^2 dt & \int e^t dt \\ \int \sin t dt & \int 3t dt \end{bmatrix} \\ &= \begin{bmatrix} \frac{t^3}{3} & e^t \\ -\cos t & \frac{3t^2}{2} \end{bmatrix} + C. \end{aligned}$$

Here  $C$  is a constant matrix of the same dimensions:

$$C = \begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}.$$

Each entry may require its own integration constant. Writing one scalar constant would incorrectly force all four constants to be equal.

### The Matrix Product Rule

If  $A(t)B(t)$  is defined for every  $t$ , then:

$$\boxed{\frac{d}{dt}[A(t)B(t)] = A'(t)B(t) + A(t)B'(t)}.$$

The order is essential. The first term is  $A'B$ , and the second is  $AB'$ . Neither factor is moved past the other.

### A Complete Product-Rule Check

Let:

$$A(t) = \begin{bmatrix} t & 1 \\ 0 & t^2 \end{bmatrix}, \quad B(t) = \begin{bmatrix} 1 & t \\ t & 0 \end{bmatrix}.$$

First multiply the matrices:

$$\begin{aligned} A(t)B(t) &= \begin{bmatrix} t(1) + 1(t) & t(t) + 1(0) \\ 0(1) + t^2(t) & 0(t) + t^2(0) \end{bmatrix} \\ &= \begin{bmatrix} 2t & t^2 \\ t^3 & 0 \end{bmatrix}. \end{aligned}$$

Differentiate this product entry by entry:

$$\frac{d}{dt}[AB] = \begin{bmatrix} 2 & 2t \\ 3t^2 & 0 \end{bmatrix}.$$

Now verify the product rule directly. Differentiate each factor:

$$A'(t) = \begin{bmatrix} 1 & 0 \\ 0 & 2t \end{bmatrix}, \quad B'(t) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Calculate the first product-rule term:

$$\begin{aligned} A'B &= \begin{bmatrix} 1 & 0 \\ 0 & 2t \end{bmatrix} \begin{bmatrix} 1 & t \\ t & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ 2t^2 & 0 \end{bmatrix}. \end{aligned}$$

Calculate the second product-rule term:

$$\begin{aligned} AB' &= \begin{bmatrix} t & 1 \\ 0 & t^2 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1 & t \\ t^2 & 0 \end{bmatrix}. \end{aligned}$$

Add corresponding entries:

$$\begin{aligned} A'B + AB' &= \begin{bmatrix} 1 & t \\ 2t^2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & t \\ t^2 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 2 & 2t \\ 3t^2 & 0 \end{bmatrix}. \end{aligned}$$

This agrees with the derivative of the product calculated first.

## A Matrix Initial-Value Calculation

Suppose:

$$X'(t) = \begin{bmatrix} 2t & 1 \\ \cos t & -e^{-t} \end{bmatrix}$$

and:

$$X(0) = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}.$$

Integrate from 0 to  $t$ :

$$X(t) - X(0) = \int_0^t X'(s) ds.$$

Calculate the integral entry by entry:

$$\begin{aligned} \int_0^t X'(s) ds &= \begin{bmatrix} \int_0^t 2s ds & \int_0^t 1 ds \\ \int_0^t \cos s ds & \int_0^t -e^{-s} ds \end{bmatrix} \\ &= \begin{bmatrix} t^2 & t \\ \sin t & e^{-t} - 1 \end{bmatrix}. \end{aligned}$$

Add  $X(0)$  to both sides:

$$\begin{aligned} X(t) &= \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix} + \begin{bmatrix} t^2 & t \\ \sin t & e^{-t} - 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 + t^2 & -2 + t \\ \sin t & 2 + e^{-t} \end{bmatrix}. \end{aligned}$$



The matrix satisfying the differential equation and initial value is:

$$X(t) = \begin{bmatrix} 1 + t^2 & -2 + t \\ \sin t & 2 + e^{-t} \end{bmatrix}.$$

Check the initial value by setting  $t = 0$ :

$$X(0) = \begin{bmatrix} 1 & -2 \\ 0 & 3 \end{bmatrix}.$$

### Block 5 Summary

Matrix differentiation and integration are entrywise operations:

$$A' = [a'_{ij}], \quad \int A \, dt = \left[ \int a_{ij} \, dt \right].$$

The matrix product rule is:

$$(AB)' = A'B + AB',$$

with factor order preserved. An indefinite matrix integral requires a constant matrix, not merely one scalar constant.

## Determinants, Eigenvalues, And Eigenvectors

### The Goal Of This Block

The goal is to build the characteristic polynomial from a determinant and then interpret its roots as special scaling factors of the matrix.

### The Two-By-Two Determinant

A determinant is a scalar associated with a square matrix.

For:

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is:

$$\det(M) = ad - bc.$$

The products use opposite corners:

main-diagonal product – other-diagonal product.

For example, let:

$$M = \begin{bmatrix} 3 & -2 \\ 5 & 1 \end{bmatrix}.$$

Substitute the entries into  $ad - bc$ :

$$\begin{aligned} \det(M) &= 3(1) - (-2)(5) \\ &= 3 - (-10) \\ &= 13. \end{aligned}$$

For this matrix, the determinant is:

$$\det(M) = 13.$$

### A Three-By-Three Determinant

For:

$$M = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix},$$

expansion along the first row gives:

$$\det(M) = a(ei - fh) - b(di - fg) + c(dh - eg).$$

The cofactor signs across the first row are:

$$+, \quad -, \quad +.$$

Consider:

$$M = \begin{bmatrix} 2 & 1 & 0 \\ -1 & 3 & 2 \\ 4 & 0 & 1 \end{bmatrix}.$$

Substitute the first-row entries and their  $2 \times 2$  determinants:

$$\begin{aligned} \det(M) &= 2[3(1) - 2(0)] - 1[(-1)(1) - 2(4)] \\ &\quad + 0[(-1)(0) - 3(4)]. \end{aligned}$$

Simplify each bracket:

$$\begin{aligned}\det(M) &= 2(3) - 1(-1 - 8) + 0 \\ &= 6 - (-9) \\ &= 15.\end{aligned}$$

For this matrix, the three-by-three determinant is:

$$\boxed{\det(M) = 15}.$$

### The Characteristic Polynomial

Let  $A$  be an  $n \times n$  matrix, and let  $I$  be the  $n \times n$  identity matrix.

The characteristic polynomial used in this chapter is:

$$\boxed{p_A(\lambda) = \det(\lambda I - A)}.$$

Here  $\lambda$  is a scalar variable. The matrix  $\lambda I - A$  is formed first, and its determinant then produces an ordinary polynomial in  $\lambda$ .

An **eigenvalue** of  $A$  is a root of that polynomial:

$$p_A(\lambda) = 0.$$

Some books define the characteristic polynomial as  $\det(A - \lambda I)$ . That convention differs by a factor  $(-1)^n$  but gives the same eigenvalues. Mixing the two conventions halfway through a calculation causes unnecessary sign errors.

## A Complete Two-By-Two Eigenvalue Example

Let:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

Start with the identity matrix:

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Multiply it by  $\lambda$ :

$$\lambda I = \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}.$$

Subtract  $A$  entry by entry:

$$\lambda I - A = \begin{bmatrix} \lambda - 4 & -2 \\ -1 & \lambda - 3 \end{bmatrix}.$$

Now take the determinant:

$$\begin{aligned} p_A(\lambda) &= \det(\lambda I - A) \\ &= (\lambda - 4)(\lambda - 3) - (-2)(-1). \end{aligned}$$

Multiply the off-diagonal entries:

$$(-2)(-1) = 2.$$

Substituting this off-diagonal product into the determinant gives:

$$p_A(\lambda) = (\lambda - 4)(\lambda - 3) - 2.$$

Expand the brackets:

$$\begin{aligned} p_A(\lambda) &= \lambda^2 - 7\lambda + 12 - 2 \\ &= \lambda^2 - 7\lambda + 10. \end{aligned}$$

Factor the polynomial:

$$p_A(\lambda) = (\lambda - 5)(\lambda - 2).$$

Set the characteristic polynomial equal to zero:

$$(\lambda - 5)(\lambda - 2) = 0.$$

Hence the eigenvalues are:

$$\boxed{\lambda_1 = 5, \quad \lambda_2 = 2}.$$

### What An Eigenvector Adds

A nonzero vector  $\mathbf{v}$  is an eigenvector associated with  $\lambda$  when:

$$\boxed{A\mathbf{v} = \lambda\mathbf{v}}.$$

The matrix may change the vector's length or direction sign, but it does not turn the vector away from its line.

To find an eigenvector, rewrite the equation as:

$$(A - \lambda I)\mathbf{v} = 0.$$

The requirement  $\mathbf{v} \neq 0$  matters. The zero vector satisfies  $A\mathbf{0} = \lambda\mathbf{0}$  for every  $\lambda$ , so it cannot distinguish one eigenvalue from another.

### Finding The Eigenvectors In The Example

Continue with:

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}.$$

For  $\lambda = 5$ :

$$A - 5I = \begin{bmatrix} -1 & 2 \\ 1 & -2 \end{bmatrix}.$$

Let:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Then  $(A - 5I)\mathbf{v} = 0$  gives:

$$-v_1 + 2v_2 = 0.$$

Add  $v_1$  to both sides:

$$2v_2 = v_1.$$

Choose  $v_2 = 1$ . Then  $v_1 = 2$ , so one eigenvector is:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Verify it locally:

$$\begin{aligned} A\mathbf{v}_1 &= \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} +10 \\ 5 \end{bmatrix} \\ &= 5 \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \end{aligned}$$

For  $\lambda = 2$ :

$$A - 2I = \begin{bmatrix} 2 & 2 \\ 1 & 1 \end{bmatrix}.$$

The equation  $(A - 2I)\mathbf{v} = 0$  reduces to:

$$v_1 + v_2 = 0.$$

Choose  $v_1 = 1$ . Then  $v_2 = -1$ , so one eigenvector is:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$



Verify:

$$\begin{aligned} A\mathbf{v}_2 &= \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ -2 \end{bmatrix} \\ &= 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}. \end{aligned}$$

Any nonzero scalar multiple of an eigenvector is another eigenvector for the same eigenvalue.

### Why Eigenvalues Matter For Differential Equations

Consider the future matrix differential equation:

$$\mathbf{x}' = A\mathbf{x}.$$

Suppose  $A\mathbf{v} = \lambda\mathbf{v}$ . Try:

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}.$$

Differentiate:

$$\mathbf{x}'(t) = \lambda e^{\lambda t}\mathbf{v}.$$

Now multiply  $\mathbf{x}$  by  $A$ :

$$\begin{aligned} A\mathbf{x}(t) &= A(e^{\lambda t}\mathbf{v}) \\ &= e^{\lambda t}A\mathbf{v} \\ &= e^{\lambda t}\lambda\mathbf{v} \\ &= \lambda e^{\lambda t}\mathbf{v}. \end{aligned}$$

Both  $\mathbf{x}'$  and  $A\mathbf{x}$  equal  $\lambda e^{\lambda t}\mathbf{v}$ . Hence the proposed function satisfies:

$$\mathbf{x}' = A\mathbf{x}.$$

The key intuition:

**an eigenvector supplies a fixed direction, while its eigenvalue controls the exponential behavior along that direction.**

A later chapter will develop this method fully.

### Triangular Matrices And Multiplicity

For a triangular matrix, the characteristic polynomial is the product of  $\lambda$  minus the diagonal entries.

Consider:

$$T = \begin{bmatrix} 3 & 1 & -2 \\ 0 & -1 & 4 \\ 0 & 0 & -1 \end{bmatrix}.$$

Then:

$$\lambda I - T = \begin{bmatrix} \lambda - 3 & -1 & 2 \\ 0 & \lambda + 1 & -4 \\ 0 & 0 & \lambda + 1 \end{bmatrix}.$$

This is triangular, so its determinant is the product of its diagonal entries:

$$\begin{aligned} p_T(\lambda) &= (\lambda - 3)(\lambda + 1)(\lambda + 1) \\ &= (\lambda - 3)(\lambda + 1)^2. \end{aligned}$$

The factorization  $p_T(\lambda) = (\lambda - 3)(\lambda + 1)^2$  shows that:

$$\lambda = 3 \text{ has multiplicity } 1$$

and:

$$\lambda = -1 \text{ has multiplicity } 2.$$

The exponent in the characteristic polynomial gives the **algebraic multiplicity**. It does not automatically tell us how many independent eigenvectors belong to that eigenvalue.

### Block 6 Summary

Using the monic convention:

$$p_A(\lambda) = \det(\lambda I - A).$$

The roots of  $p_A(\lambda) = 0$  are the eigenvalues. An associated nonzero vector satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

Eigenpairs matter in ODEs because they generate modes of the form  $e^{\lambda t}\mathbf{v}$ .

## Cayley-Hamilton And A Reliable Matrix Workflow

### The Goal Of This Block

The goal is to understand what it means for a matrix to satisfy its own characteristic equation and to use that identity to reduce high powers.

### Substituting A Matrix Into A Polynomial

Suppose:

$$p(\lambda) = \lambda^2 - 4\lambda + 7.$$

Replacing the scalar variable  $\lambda$  by a square matrix  $A$  gives:

$$p(A) = A^2 - 4A + 7I.$$

The constant term becomes  $7I$ , not the scalar  $7$ , because every term in the sum must be a matrix of the same dimensions.

More generally, if:

$$p(\lambda) = \lambda^n + c_{n-1}\lambda^{n-1} + \cdots + c_1\lambda + c_0,$$

then:

$$p(A) = A^n + c_{n-1}A^{n-1} + \cdots + c_1A + c_0I.$$

## The Cayley-Hamilton Theorem

Let  $p_A(\lambda)$  be the characteristic polynomial of a square matrix  $A$ . The Cayley-Hamilton theorem states:

$$p_A(A) = 0.$$

The zero on the right is the zero matrix of the same dimensions as  $A$ .

This does not say that  $p_A(\lambda) = 0$  for every scalar  $\lambda$ . It says that the matrix expression obtained by replacing  $\lambda$  with  $A$  equals the zero matrix.

## A Complete Cayley-Hamilton Verification

Let:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}.$$

Construct  $\lambda I - A$ :

$$\lambda I - A = \begin{bmatrix} \lambda - 1 & -2 \\ -3 & \lambda \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned} p_A(\lambda) &= (\lambda - 1)\lambda - (-2)(-3) \\ &= \lambda^2 - \lambda - 6. \end{aligned}$$

Therefore the characteristic equation is:

$$\lambda^2 - \lambda - 6 = 0.$$

Cayley-Hamilton predicts:

$$A^2 - A - 6I = 0.$$

Calculate  $A^2$  from the matrix product  $AA$ :

$$\begin{aligned} A^2 &= \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 1(1) + 2(3) & 1(2) + 2(0) \\ 3(1) + 0(3) & 3(2) + 0(0) \end{bmatrix} \\ &= \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix}. \end{aligned}$$

Now subtract  $A$ :

$$\begin{aligned} A^2 - A &= \begin{bmatrix} 7 & 2 \\ 3 & 6 \end{bmatrix} - \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} \\ &= \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}. \end{aligned}$$

Write  $6I$  explicitly:

$$6I = \begin{bmatrix} 6 & 0 \\ 0 & 6 \end{bmatrix}.$$

The two displayed matrices are equal, so:

$$A^2 - A = 6I.$$

Subtract  $6I$  from both sides:

$$\boxed{A^2 - A - 6I = 0}.$$

This verifies the theorem for the chosen matrix.

### Reducing A High Matrix Power

The verified identity is:

$$A^2 = A + 6I.$$

Use it to find  $A^5$  without performing four full matrix multiplications.

Multiply the identity by  $A$ :

$$A^3 = A(A + 6I).$$

Distribute:

$$A^3 = A^2 + 6A.$$

Replace  $A^2$  by  $A + 6I$ :

$$\begin{aligned} A^3 &= (A + 6I) + 6A \\ &= 7A + 6I. \end{aligned}$$

Multiply by  $A$  again:

$$A^4 = A(7A + 6I).$$

Distribute:

$$A^4 = 7A^2 + 6A.$$

Replace  $A^2$ :

$$\begin{aligned} A^4 &= 7(A + 6I) + 6A \\ &= 7A + 42I + 6A \\ &= 13A + 42I. \end{aligned}$$

Multiply by  $A$  once more:

$$A^5 = A(13A + 42I).$$

Distribute:

$$A^5 = 13A^2 + 42A.$$

Replace  $A^2$  by  $A + 6I$ :

$$\begin{aligned} A^5 &= 13(A + 6I) + 42A \\ &= 13A + 78I + 42A \\ &= 55A + 78I. \end{aligned}$$

Now insert:

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$



Then:

$$\begin{aligned} A^5 &= 55 \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix} + 78 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 55 & 110 \\ 165 & 0 \end{bmatrix} + \begin{bmatrix} 78 & 0 \\ 0 & 78 \end{bmatrix} \\ &= \begin{bmatrix} 133 & 110 \\ 165 & 78 \end{bmatrix}. \end{aligned}$$

Therefore:

$$A^5 = \begin{bmatrix} 133 & 110 \\ 165 & 78 \end{bmatrix}.$$

The theorem converts every power  $A^n$  with  $n \geq 2$  into a linear combination of  $A$  and  $I$  for this  $2 \times 2$  example.

### A Reliable Workflow

**Step 1: Write dimensions beside every matrix.** Do this before adding or multiplying.

**Step 2: Test compatibility.** Addition needs identical dimensions. Multiplication needs matching inner dimensions.

**Step 3: Predict the output dimensions.** For  $(m \times n)(n \times p)$ , create an empty  $m \times p$  result before doing arithmetic.

**Step 4: Name the row and column for each product entry.** This prevents accidental entrywise multiplication.

**Step 5: Preserve factor order.** Never commute factors unless commutativity has been proved for those matrices.

**Step 6: For eigenvalues, construct  $\lambda I - A$  explicitly.** Then take the determinant and solve the resulting scalar polynomial.

**Step 7: For an eigenvector, solve  $(A - \lambda I)\mathbf{v} = 0$ .** Keep the requirement  $\mathbf{v} \neq 0$  visible.

**Step 8: For Cayley-Hamilton, convert constants to multiples of  $I$ .** Then verify or reduce powers one substitution at a time.

### Common Mistake: Multiplying Corresponding Entries

Matrix multiplication is not usually entrywise multiplication.

For  $AB$ , the entry  $(AB)_{ij}$  comes from:

row  $i$  of  $A$    with   column  $j$  of  $B$ .

If corresponding entries are multiplied instead, a different operation has been performed.

### Common Mistake: Checking The Outer Dimensions

For:

$$(m \times n)(r \times p),$$

the product exists only when:

$$\boxed{n = r}.$$

The inner dimensions must match. The outer dimensions  $m$  and  $p$  describe the answer; they do not decide whether multiplication is legal.

**Common Mistake: Reversing A Product**

From  $AB$ , one cannot casually move to  $BA$ .

Possible outcomes include:

- both products exist but have different values
- both products exist but have different interpretations
- one product exists while the other does not
- both products exist but have different dimensions

Always retest dimensions after reversing the order.

**Common Mistake: Losing Constants In A Matrix Integral**

If:

$$X'(t) = A(t),$$

then an indefinite integration gives:

$$X(t) = \int A(t) dt + C,$$

where  $C$  is a constant matrix. Each entry can have an independent constant.

**Common Mistake: Treating An Eigenvalue As A Vector**

An eigenvalue  $\lambda$  is a scalar. An eigenvector  $\mathbf{v}$  is a nonzero vector. They are linked by:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The equation does not say  $A\lambda = \mathbf{v}$  or  $A\mathbf{v} = \mathbf{v}$  unless  $\lambda = 1$ .

**Common Mistake: Reading Multiplicity From A Matrix Entry**

Algebraic multiplicity comes from the exponent of a factor in the characteristic polynomial.

If:

$$p_A(\lambda) = (\lambda - 4)^3(\lambda + 2),$$

then  $\lambda = 4$  has algebraic multiplicity 3. Repeated appearances of the number 4 among the matrix entries would not establish that fact.

**Common Mistake: Substituting A Scalar Constant Into Cayley-Hamilton**

If:

$$p_A(\lambda) = \lambda^2 - 5\lambda + 6,$$

then:

$$p_A(A) = A^2 - 5A + 6I.$$

The final term is  $6I$ , because it must be a matrix. Writing  $A^2 - 5A + 6$  tries to add a scalar directly to matrices.

**Block 7 Summary**

The Cayley-Hamilton theorem states:

$$p_A(A) = 0.$$

It turns the characteristic polynomial into a matrix identity. That identity can verify a calculation, reduce high powers, and later simplify functions of matrices. Reliable work begins with dimensions, preserves factor order, and keeps scalar, vector, and matrix objects distinct.

## Original Practice Set

### Practice Problems

**Problem 1: Read Entries And Form A Transpose.** Suppose:

$$A = \begin{bmatrix} 2 & a & -1 \\ b & 4 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 5 & -1 \\ -3 & 4 & 0 \end{bmatrix}.$$

Find  $a$  and  $b$ , state the dimensions of  $A$ , and write  $A^T$ .

**Problem 2: Calculate A Linear Combination.** Let:

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} -2 & 5 & 1 \\ 3 & -1 & 2 \end{bmatrix}.$$

Find  $2A - 3B$ .

**Problem 3: Test Product Dimensions.** Suppose:

$$A : 3 \times 2, \quad B : 2 \times 4, \quad C : 4 \times 3.$$

For each expression, state whether it is defined. If it is defined, state its dimensions:

$$AB, \quad BA, \quad BC, \quad CA, \quad AC, \quad (AB)C.$$

**Problem 4: Calculate A Rectangular Product.** Let:

$$P = \begin{bmatrix} 2 & -1 & 3 \\ 0 & 4 & 1 \end{bmatrix}, \quad Q = \begin{bmatrix} 1 & 2 \\ -2 & 0 \\ 3 & -1 \end{bmatrix}.$$

Find  $PQ$ . Show the row-column calculation for every entry.

**Problem 5: Compare Both Product Orders.** Let:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix}.$$

Find  $AB$  and  $BA$ . Use the results to decide whether these matrices commute.

**Problem 6: Multiply A Matrix By A Vector.** Let:

$$M = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 3 & 2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}.$$

Find  $M\mathbf{x}$  by row-column multiplication. Then reproduce the result as a linear combination of the columns of  $M$ .

**Problem 7: Calculate Matrix Powers.** Let:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

Find  $A^2$  and  $A^3$ . Explain briefly why squaring each entry would not produce  $A^2$ .

**Problem 8: Differentiate And Integrate A Matrix.** Let:

$$A(t) = \begin{bmatrix} t^3 & \cos t \\ e^{-t} & 2t - 1 \end{bmatrix}.$$

Find:

$$A'(t)$$

and:

$$\int_0^1 A(t) dt.$$

**Problem 9: Use The Matrix Product Rule.** Let:

$$A(t) = \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix}, \quad B(t) = \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix}.$$

Use:

$$(AB)' = A'B + AB'$$

to find  $\frac{d}{dt}[A(t)B(t)]$ . Then check the result by multiplying  $AB$  first and differentiating afterward.

**Problem 10: Find Eigenvalues And Eigenvectors.** Let:

$$C = \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix}.$$

Find the characteristic polynomial, both eigenvalues, and one eigenvector for each eigenvalue.

**Problem 11: Read Multiplicity From A Triangular Matrix.** Let:

$$T = \begin{bmatrix} 4 & 2 & 0 \\ 0 & 4 & -1 \\ 0 & 0 & -3 \end{bmatrix}.$$

Find the characteristic polynomial and list each eigenvalue with its algebraic multiplicity.

**Problem 12: Verify Cayley-Hamilton And Reduce A Power.** Let:

$$D = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix}.$$

1. Find  $p_D(\lambda) = \det(\lambda I - D)$ .
2. Verify  $p_D(D) = 0$ .
3. Use the resulting matrix identity to find  $D^4$ .



**Worked Answers: Problems 1 To 4**

**Answer 1.** Two matrices are equal when corresponding entries are equal.

Compare the row 1, column 2 entries:

$$a = 5.$$

Compare the row 2, column 1 entries:

$$b = -3.$$

The matrix has two rows and three columns, so:

$$\boxed{A \text{ is } 2 \times 3}.$$

Substitute  $a = 5$  and  $b = -3$ :

$$A = \begin{bmatrix} 2 & 5 & -1 \\ -3 & 4 & 0 \end{bmatrix}.$$

To transpose  $A$ , turn each row into a column. The first row becomes:

$$\begin{bmatrix} 2 \\ 5 \\ -1 \end{bmatrix},$$

and the second row becomes:

$$\begin{bmatrix} -3 \\ 4 \\ 0 \end{bmatrix}.$$

Placing the two row vectors of  $A$  into columns gives:

$$A^T = \begin{bmatrix} 2 & -3 \\ 5 & 4 \\ -1 & 0 \end{bmatrix}.$$

The transpose has dimensions  $3 \times 2$ .

**Answer 2.** The requested expression is:

$$2A - 3B.$$

First multiply every entry of  $A$  by 2:

$$\begin{aligned} 2A &= 2 \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 2 & -4 & 6 \\ 8 & 0 & -2 \end{bmatrix}. \end{aligned}$$

Now multiply every entry of  $B$  by 3:

$$\begin{aligned} 3B &= 3 \begin{bmatrix} -2 & 5 & 1 \\ 3 & -1 & 2 \end{bmatrix} \\ &= \begin{bmatrix} -6 & 15 & 3 \\ 9 & -3 & 6 \end{bmatrix}. \end{aligned}$$

Subtract corresponding entries:

$$\begin{aligned} 2A - 3B &= \begin{bmatrix} 2 - (-6) & -4 - 15 & 6 - 3 \\ 8 - 9 & 0 - (-3) & -2 - 6 \end{bmatrix} \\ &= \begin{bmatrix} 8 & -19 & 3 \\ -1 & 3 & -8 \end{bmatrix}. \end{aligned}$$

The completed calculation of  $2A - 3B$  gives:

$$2A - 3B = \begin{bmatrix} 8 & -19 & 3 \\ -1 & 3 & -8 \end{bmatrix}.$$

**Answer 3.** A product:

$$(m \times n)(r \times p)$$

is defined only when the inner dimensions match:

$$n = r.$$

When it is defined, the answer has dimensions  $m \times p$ .

For  $AB$ :

$$(3 \times 2)(2 \times 4).$$

The inner dimensions are both 2, so:

$$AB \text{ is defined and is } 3 \times 4.$$

For  $BA$ :

$$(2 \times 4)(3 \times 2).$$

The inner dimensions are 4 and 3, so:

$$BA \text{ is not defined}.$$

For  $BC$ :

$$(2 \times 4)(4 \times 3).$$

The inner dimensions are both 4, so:

$$BC \text{ is defined and is } 2 \times 3.$$

For  $CA$ :

$$(4 \times 3)(3 \times 2).$$

The inner dimensions are both 3, so:

$$CA \text{ is defined and is } 4 \times 2.$$

For  $AC$ :

$$(3 \times 2)(4 \times 3).$$

The inner dimensions are 2 and 4, so:

$$AC \text{ is not defined}.$$

Finally,  $AB$  has dimensions  $3 \times 4$ . Therefore:

$$(AB)C \text{ has dimensions } (3 \times 4)(4 \times 3).$$

The inner dimensions match, so:

$$(AB)C \text{ is defined and is } 3 \times 3.$$

**Answer 4.** The dimensions are:

$$P : 2 \times 3, \quad Q : 3 \times 2.$$

Therefore  $PQ$  is defined and will be  $2 \times 2$ .

For row 1, column 1:

$$\begin{aligned}(PQ)_{11} &= 2(1) + (-1)(-2) + 3(3) \\ &= 2 + 2 + 9 \\ &= 13.\end{aligned}$$

For row 1, column 2:

$$\begin{aligned}(PQ)_{12} &= 2(2) + (-1)(0) + 3(-1) \\ &= 4 + 0 - 3 \\ &= 1.\end{aligned}$$

For row 2, column 1:

$$\begin{aligned}(PQ)_{21} &= 0(1) + 4(-2) + 1(3) \\ &= 0 - 8 + 3 \\ &= -5.\end{aligned}$$

For row 2, column 2:

$$\begin{aligned}(PQ)_{22} &= 0(2) + 4(0) + 1(-1) \\ &= 0 + 0 - 1 \\ &= -1.\end{aligned}$$

Place these entries in their corresponding positions:

$$PQ = \begin{bmatrix} 13 & 1 \\ -5 & -1 \end{bmatrix}.$$

**Worked Answers: Problems 5 To 8**

**Answer 5.** Both matrices are  $2 \times 2$ , so both  $AB$  and  $BA$  are defined.

Calculate  $AB$ :

$$\begin{aligned} AB &= \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 1(3) + 2(4) & 1(0) + 2(2) \\ 0(3) + (-1)(4) & 0(0) + (-1)(2) \end{bmatrix} \\ &= \begin{bmatrix} 11 & 4 \\ -4 & -2 \end{bmatrix}. \end{aligned}$$

Now reverse the order:

$$\begin{aligned} BA &= \begin{bmatrix} 3 & 0 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} 3(1) + 0(0) & 3(2) + 0(-1) \\ 4(1) + 2(0) & 4(2) + 2(-1) \end{bmatrix} \\ &= \begin{bmatrix} 3 & 6 \\ 4 & 6 \end{bmatrix}. \end{aligned}$$

Therefore:

$$AB = \begin{bmatrix} 11 & 4 \\ -4 & -2 \end{bmatrix} \neq \begin{bmatrix} 3 & 6 \\ 4 & 6 \end{bmatrix} = BA.$$

The matrices do not commute.

**Answer 6.** The dimensions are:

$$M : 2 \times 3, \quad \mathbf{x} : 3 \times 1.$$

Therefore  $M\mathbf{x}$  is defined and will be  $2 \times 1$ .

Use the first row of  $M$  for the first output entry:

$$\begin{aligned}(M\mathbf{x})_1 &= 2(1) + 1(-2) + (-1)(4) \\ &= 2 - 2 - 4 \\ &= -4.\end{aligned}$$

Use the second row for the second output entry:

$$\begin{aligned}(M\mathbf{x})_2 &= 0(1) + 3(-2) + 2(4) \\ &= 0 - 6 + 8 \\ &= 2.\end{aligned}$$

The row-column calculation gives:

$$M\mathbf{x} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}.$$

Now write the columns of  $M$  as:

$$\mathbf{m}_1 = \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \quad \mathbf{m}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \mathbf{m}_3 = \begin{bmatrix} -1 \\ 2 \end{bmatrix}.$$



The entries of  $\mathbf{x}$  are 1,  $-2$ , and 4. Therefore:

$$M\mathbf{x} = 1\mathbf{m}_1 - 2\mathbf{m}_2 + 4\mathbf{m}_3.$$

Substitute the columns:

$$\begin{aligned} M\mathbf{x} &= 1 \begin{bmatrix} 2 \\ 0 \end{bmatrix} - 2 \begin{bmatrix} 1 \\ 3 \end{bmatrix} + 4 \begin{bmatrix} -1 \\ 2 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 0 \end{bmatrix} + \begin{bmatrix} -2 \\ -6 \end{bmatrix} + \begin{bmatrix} -4 \\ 8 \end{bmatrix} \\ &= \begin{bmatrix} -4 \\ 2 \end{bmatrix}. \end{aligned}$$

The column-combination interpretation gives the same result.

**Answer 7.** A matrix power uses matrix multiplication:

$$A^2 = AA.$$

Substitute:

$$A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

Then:

$$\begin{aligned} A^2 &= \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 2(2) + 1(0) & 2(1) + 1(2) \\ 0(2) + 2(0) & 0(1) + 2(2) \end{bmatrix} \\ &= \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}. \end{aligned}$$

The matrix product  $AA$  gives:

$$A^2 = \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix}.$$

Now use:

$$A^3 = A^2A.$$

Substitute the calculated  $A^2$  and the original  $A$ :

$$\begin{aligned} A^3 &= \begin{bmatrix} 4 & 4 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \\ &= \begin{bmatrix} 4(2) + 4(0) & 4(1) + 4(2) \\ 0(2) + 4(0) & 0(1) + 4(2) \end{bmatrix} \\ &= \begin{bmatrix} 8 & 12 \\ 0 & 8 \end{bmatrix}. \end{aligned}$$

Therefore:

$$A^3 = \begin{bmatrix} 8 & 12 \\ 0 & 8 \end{bmatrix}.$$

Entrywise squaring would produce:

$$\begin{bmatrix} 4 & 1 \\ 0 & 4 \end{bmatrix},$$

which differs from  $A^2$ . The notation  $A^2$  means  $AA$ .

**Answer 8.** Differentiate each entry of:

$$A(t) = \begin{bmatrix} t^3 & \cos t \\ e^{-t} & 2t - 1 \end{bmatrix}.$$

The entrywise derivative is:

$$\begin{aligned} A'(t) &= \begin{bmatrix} \frac{d}{dt}(t^3) & \frac{d}{dt}(\cos t) \\ \frac{d}{dt}(e^{-t}) & \frac{d}{dt}(2t - 1) \end{bmatrix} \\ &= \begin{bmatrix} 3t^2 & -\sin t \\ -e^{-t} & 2 \end{bmatrix}. \end{aligned}$$

Thus:

$$A'(t) = \begin{bmatrix} 3t^2 & -\sin t \\ -e^{-t} & 2 \end{bmatrix}.$$

For the definite integral, integrate each entry from 0 to 1:

$$\int_0^1 A(t) dt = \begin{bmatrix} \int_0^1 t^3 dt & \int_0^1 \cos t dt \\ \int_0^1 e^{-t} dt & \int_0^1 (2t - 1) dt \end{bmatrix}.$$

Calculate the top-left entry:

$$\begin{aligned} \int_0^1 t^3 dt &= \left[ \frac{t^4}{4} \right]_0^1 \\ &= \frac{1}{4}. \end{aligned}$$

Calculate the top-right entry:

$$\begin{aligned}\int_0^1 \cos t \, dt &= [\sin t]_0^1 \\ &= \sin 1.\end{aligned}$$

Calculate the bottom-left entry:

$$\begin{aligned}\int_0^1 e^{-t} \, dt &= [-e^{-t}]_0^1 \\ &= -e^{-1} - (-1) \\ &= 1 - e^{-1}.\end{aligned}$$

Calculate the bottom-right entry:

$$\begin{aligned}\int_0^1 (2t - 1) \, dt &= [t^2 - t]_0^1 \\ &= (1 - 1) - (0 - 0) \\ &= 0.\end{aligned}$$

Place the four results in their original positions:

$$\int_0^1 A(t) \, dt = \begin{bmatrix} \frac{1}{4} & \sin 1 \\ 1 - e^{-1} & 0 \end{bmatrix}.$$

A definite integral does not require an arbitrary constant matrix.

**Worked Answers: Problems 9 To 12**

**Answer 9.** The product rule is:

$$(AB)' = A'B + AB'.$$

Differentiate  $A(t)$  entry by entry:

$$A'(t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Differentiate  $B(t)$  entry by entry:

$$B'(t) = \begin{bmatrix} e^t & 0 \\ 1 & 0 \end{bmatrix}.$$

Calculate the first product-rule term:

$$\begin{aligned} A'B &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} \\ &= \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix}. \end{aligned}$$

Calculate the second term:

$$\begin{aligned} AB' &= \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} e^t & 0 \\ 1 & 0 \end{bmatrix} \\ &= \begin{bmatrix} te^t + 1 & 0 \\ t & 0 \end{bmatrix}. \end{aligned}$$

Add corresponding entries:

$$\begin{aligned} A'B + AB' &= \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} + \begin{bmatrix} te^t + 1 & 0 \\ t & 0 \end{bmatrix} \\ &= \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}. \end{aligned}$$

Therefore:

$$\boxed{\frac{d}{dt}[A(t)B(t)] = \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}}.$$

To check, multiply first:

$$\begin{aligned} A(t)B(t) &= \begin{bmatrix} t & 1 \\ 0 & t \end{bmatrix} \begin{bmatrix} e^t & 0 \\ t & 1 \end{bmatrix} \\ &= \begin{bmatrix} te^t + t & 1 \\ t^2 & t \end{bmatrix}. \end{aligned}$$

Differentiate the product entry by entry:

$$\frac{d}{dt}[AB] = \begin{bmatrix} (t+1)e^t + 1 & 0 \\ 2t & 1 \end{bmatrix}.$$

The two methods agree.

**Answer 10.** The characteristic polynomial is:

$$p_C(\lambda) = \det(\lambda I - C).$$

Construct the matrix:

$$\lambda I - C = \begin{bmatrix} \lambda - 5 & -2 \\ -2 & \lambda - 2 \end{bmatrix}.$$

Take its determinant:

$$\begin{aligned} p_C(\lambda) &= (\lambda - 5)(\lambda - 2) - (-2)(-2) \\ &= (\lambda - 5)(\lambda - 2) - 4. \end{aligned}$$

Expand:

$$\begin{aligned} p_C(\lambda) &= \lambda^2 - 7\lambda + 10 - 4 \\ &= \lambda^2 - 7\lambda + 6. \end{aligned}$$

Factor:

$$p_C(\lambda) = (\lambda - 6)(\lambda - 1).$$

Therefore:

$$\boxed{\lambda_1 = 6, \quad \lambda_2 = 1}.$$

For  $\lambda = 6$ , solve:

$$(C - 6I)\mathbf{v} = 0.$$

The coefficient matrix is:

$$C - 6I = \begin{bmatrix} -1 & 2 \\ 2 & -4 \end{bmatrix}.$$

The first row gives:

$$-v_1 + 2v_2 = 0.$$

Hence:

$$v_1 = 2v_2.$$

Choose  $v_2 = 1$ , which gives  $v_1 = 2$ . One eigenvector is:

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Check:

$$\begin{aligned} C\mathbf{v}_1 &= \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 12 \\ 6 \end{bmatrix} \\ &= 6 \begin{bmatrix} 2 \\ 1 \end{bmatrix}. \end{aligned}$$

For  $\lambda = 1$ , solve:

$$(C - I)\mathbf{v} = 0.$$

The coefficient matrix is:

$$C - I = \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}.$$



The second row gives:

$$2v_1 + v_2 = 0.$$

Therefore:

$$v_2 = -2v_1.$$

Choose  $v_1 = 1$ , so one eigenvector is:

$$\mathbf{v}_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}.$$

Check:

$$\begin{aligned} C\mathbf{v}_2 &= \begin{bmatrix} 5 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= \begin{bmatrix} 1 \\ -2 \end{bmatrix} \\ &= 1 \begin{bmatrix} 1 \\ -2 \end{bmatrix}. \end{aligned}$$

**Answer 11.** The characteristic polynomial is:

$$p_T(\lambda) = \det(\lambda I - T).$$

Construct:

$$\lambda I - T = \begin{bmatrix} \lambda - 4 & -2 & 0 \\ 0 & \lambda - 4 & 1 \\ 0 & 0 & \lambda + 3 \end{bmatrix}.$$

This matrix is triangular. Its determinant is the product of its diagonal entries:

$$\begin{aligned}p_T(\lambda) &= (\lambda - 4)(\lambda - 4)(\lambda + 3) \\ &= (\lambda - 4)^2(\lambda + 3).\end{aligned}$$

Therefore:

$$\lambda = 4 \text{ has algebraic multiplicity } 2$$

and:

$$\lambda = -3 \text{ has algebraic multiplicity } 1.$$

The off-diagonal entries do not change the fact that a triangular matrix's eigenvalues are its diagonal entries.

**Answer 12.** The characteristic polynomial is:

$$p_D(\lambda) = \det(\lambda I - D).$$

For:

$$D = \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix},$$

construct:

$$\lambda I - D = \begin{bmatrix} \lambda & -2 \\ -1 & \lambda - 3 \end{bmatrix}.$$

Take the determinant:

$$\begin{aligned}p_D(\lambda) &= \lambda(\lambda - 3) - (-2)(-1) \\&= \lambda^2 - 3\lambda - 2.\end{aligned}$$

Thus:

$$\boxed{p_D(\lambda) = \lambda^2 - 3\lambda - 2}.$$

Cayley-Hamilton predicts:

$$D^2 - 3D - 2I = 0.$$

Calculate  $D^2$ :

$$\begin{aligned}D^2 &= \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} \\&= \begin{bmatrix} 0(0) + 2(1) & 0(2) + 2(3) \\ 1(0) + 3(1) & 1(2) + 3(3) \end{bmatrix} \\&= \begin{bmatrix} 2 & 6 \\ 3 & 11 \end{bmatrix}.\end{aligned}$$

Calculate  $3D$ :

$$3D = \begin{bmatrix} 0 & 6 \\ 3 & 9 \end{bmatrix}.$$

Subtract:

$$\begin{aligned}D^2 - 3D &= \begin{bmatrix} 2 & 6 \\ 3 & 11 \end{bmatrix} - \begin{bmatrix} 0 & 6 \\ 3 & 9 \end{bmatrix} \\&= \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} \\&= 2I.\end{aligned}$$

Subtract  $2I$  from both sides:

$$\boxed{D^2 - 3D - 2I = 0}.$$

Rearrange the verified identity:

$$D^2 = 3D + 2I.$$

Multiply both sides by  $D$ :

$$D^3 = 3D^2 + 2D.$$

Replace  $D^2$  with  $3D + 2I$ :

$$\begin{aligned} D^3 &= 3(3D + 2I) + 2D \\ &= 9D + 6I + 2D \\ &= 11D + 6I. \end{aligned}$$

Multiply by  $D$  again:

$$D^4 = 11D^2 + 6D.$$

Replace  $D^2$  once more:

$$\begin{aligned} D^4 &= 11(3D + 2I) + 6D \\ &= 33D + 22I + 6D \\ &= 39D + 22I. \end{aligned}$$

Insert  $D$  and  $I$ :

$$\begin{aligned} D^4 &= 39 \begin{bmatrix} 0 & 2 \\ 1 & 3 \end{bmatrix} + 22 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 78 \\ 39 & 117 \end{bmatrix} + \begin{bmatrix} 22 & 0 \\ 0 & 22 \end{bmatrix} \\ &= \begin{bmatrix} 22 & 78 \\ 39 & 139 \end{bmatrix}. \end{aligned}$$

Therefore:

$$D^4 = \begin{bmatrix} 22 & 78 \\ 39 & 139 \end{bmatrix}.$$

### Chapter 15 Final Summary

An  $m \times n$  matrix has  $m$  rows and  $n$  columns, and  $a_{ij}$  denotes its row  $i$ , column  $j$  entry.

The core compatibility rules are:

$$\begin{aligned} A + B &= [a_{ij} + b_{ij}] \quad \text{when the dimensions match,} \\ (m \times n)(n \times p) &\longrightarrow m \times p, \\ (AB)_{ij} &= \sum_{k=1}^n a_{ik} b_{kj}. \end{aligned}$$

Matrix multiplication is associative and distributive, but generally  $AB \neq BA$ . For square matrices and matrix-valued functions:

$$\begin{aligned} A^0 &= I, & A^n &= \underbrace{A A \cdots A}_{n \text{ factors}}, \\ A' &= [a'_{ij}], & \int A \, dt &= \left[ \int a_{ij} \, dt \right] + C, \\ (AB)' &= A'B + AB'. \end{aligned}$$

Using the monic characteristic-polynomial convention:

$$p_A(\lambda) = \det(\lambda I - A).$$

Its roots are eigenvalues. An associated nonzero eigenvector satisfies:

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The Cayley-Hamilton theorem converts that scalar polynomial into the matrix identity:

$$p_A(A) = 0.$$

This identity can reduce higher matrix powers.

The chapter's central lesson is:

**begin with dimensions, preserve factor order, and identify whether each object is a scalar, vector, or matrix before applying an algebraic rule.**